

Transition: DS + Algorithms

More
Recurrences



Defn A linear homogeneous recurrence has the form

$$a_n = C_1 a_{n-1} + C_2 a_{n-2} + \dots + C_d a_{n-d}$$

where $C_1, \dots, C_d$ are constant and $C_d \neq 0$

The order of the recurrence is d .

Examples:

$$a_n = 3a_{n-1} + 2a_{n-2} \rightarrow \text{order 2}$$

$$T(n) = 0 \cdot T(n-1) + 0 \cdot T(n-2) + 0 \cdot T(n-3) + 4 \cdot T(n-4) \quad T(n) = 4T(n-4) \quad \text{order 4}$$

$$F_n = F_{n-1} + F_{n-2} \quad \text{order 2}$$

Examples: Yes or no? Is it linear hom rec?

$$P(n) = 1.06 P(n-1)$$

yes $\rightarrow$ order 1

$$H(n) = 2H(n-1) + 1$$

no $\rightarrow$ added fn

$$R(n) = R(n-1) + (R(n-2))^2 \rightarrow \text{no, not linear}$$

$$a_n = \sqrt{3} a_{n-5}$$

yes $c_5 = \sqrt{3}$
order 5 c_1, c_2, c_3, c_4
all = 0

$$f_n = n \cdot f_{n-1}$$

n is not constant

$$M(n) = 2M\left(\frac{n}{2}\right) + n$$

no

$\uparrow$ term
 $n/2$ is not $n-d$

How to solve:

Generally, these types have solutions which are exponential.

So $\rightarrow$ look for equations of the

$$\text{form } a_n = r^n$$

where r is some constant

If a solution of this form exists,

$$\text{have: } a_n = C_1 a_{n-1} + C_2 a_{n-2} + \dots + C_d a_{n-d}$$

$$\Downarrow$$
$$r^n = C_1 r^{n-1} + C_2 r^{n-2} + \dots + C_d r^{n-d}$$

meth!

divide common term:

$$r^{n-d}$$

$$\Downarrow$$
$$r^d = C_1 r^{d-1} + C_2 r^{d-2} + \dots + C_d$$

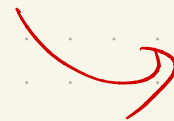
So: The sequence $a_n = r^n$ is a solution



r is a solution of the characteristic eqn:

$$\underline{r^d} - c_1 r^{d-1} - \dots - c_d = 0$$

How to find r ? math!



Worksheet:

Ex: $B(n) = 2B(n-1) + B(n-2)$

... This is a bit different!

Finding general solutions

- If r is a non-repeated root of the char. eqn, then r^n is a solution to the recurrence.
- If r is a repeated root with multiplicity k then $r, n \cdot r^n, \dots, n^{k-1} r^n$ are all solutions
- Any linear combination will satisfy the recurrence, but need to use initial values to find constants

Huh?

Examples

- $P(n) = 1.06 P(n-1)$

$$P(0) = 10,000$$

$$x - 1.06 = 0 \Rightarrow \text{root is } x = 1.06$$

so $P(n) = C \cdot (1.06)^n$

what is C ?

$$\bullet A(n) = A(n-1) + 2A(n-2)$$

$$\bullet A(0) =$$

$$A(1) =$$

$$\Rightarrow x^2 - x - 2 = 0 \text{ had roots}$$

$$x = 2 \text{ and } x = -1$$

$$\text{So: } A(n) = C_1 \cdot 2^n + C_2 \cdot (-1)^n$$

- Suppose a characteristic equation for $C(n)$ simplifies to:

$$(x-2)^3 (x-5)^2 (x+9) = 0$$

Then roots are:

$$\text{and } C(n) \geq$$

So:

We can find runtimes, as long as
the recurrence is linear &
homogeneous!

Q: What if it's not??

Two others we'll study here,
since they are useful in computing.

- linear inhomogeneous

- divide & conquer

Linear inhomogeneous recurrences

These are just linear ones, with an added function $g(n)$ at end:

$$f(n) = c_1 f(n-1) + \dots + c_d f(n-d) + g(n)$$

Examples:

$$F(n) = F(n-1) + F(n-2) + 1$$

$$A(n) = 4A(n-1) - 2^n + n^2$$

Next time: how to solve?

Method to solve

① "Ignore" $g(n)$, & find the general solution for the linear homogeneous part. How?

② Find a general solution for $g(n)$
↳ "particular solution"

③ Add them together

④ Use base cases to solve for constants.

So: Step 2 is the critical unknown.

Depending on $g(n)$, this can be hard.

But: If $g(n) = [\text{polynomial}] \times 5^n$
 $\uparrow$ some constant

then can do it.

Ex: $g(n) = (n^2 + 1) \cdot 5^n$

or $g(n) = 5n - 2$

Process

Is s a char root?

No
↙

try a general
solution of the
form:
 $s^n \cdot (\text{poly of deg } k)$

new
constants!

yes
↓

What is its
multiplicity?
Let this
be m

Try a general solution
of the form

$t^m \cdot (\text{poly. of deg } k) \cdot s^n$

Example

$$f(0) = 1$$

$$f(n) = 4f(n-1) + 3^n$$

Step 1

Step 2

Step 3:

Step 4:

Another: $a_n = 3a_{n-1} + 4a_{n-2} + n \cdot 4^n$

Next reading: Divide & conquer

MERGESORT($A[1..n]$):

if $n > 1$

$m \leftarrow \lfloor n/2 \rfloor$

MERGESORT($A[1..m]$) *⟨⟨Recurse!⟩⟩*

MERGESORT($A[m+1..n]$) *⟨⟨Recurse!⟩⟩*

MERGE($A[1..n], m$)

MERGE($A[1..n], m$):

$i \leftarrow 1; j \leftarrow m+1$

for $k \leftarrow 1$ to n

if $j > n$

$B[k] \leftarrow A[i]; i \leftarrow i+1$

else if $i > m$

$B[k] \leftarrow A[j]; j \leftarrow j+1$

else if $A[i] < A[j]$

$B[k] \leftarrow A[i]; i \leftarrow i+1$

else

$B[k] \leftarrow A[j]; j \leftarrow j+1$

for $k \leftarrow 1$ to n

$A[k] \leftarrow B[k]$

Figure 1.6. Mergesort

Mergesort: 2 parts

Part 1: Merge works

Setup: Given $A[1..n]$ and an index m
with $1 \leq m \leq n$ where $A[1..m]$ +
 $A[m+1..n]$ are sorted, MERGE
correctly sorts $A[1..n]$ by end.

How?

Part 2:

Induction of length of $A[1..n]$!

Runtime:

MERGESORT($A[1..n]$):

if $n > 1$

$m \leftarrow \lfloor n/2 \rfloor$

MERGESORT($A[1..m]$) *⟨⟨Recurse!⟩⟩*

MERGESORT($A[m+1..n]$) *⟨⟨Recurse!⟩⟩*

MERGE($A[1..n], m$)

MERGE($A[1..n], m$):

$i \leftarrow 1; j \leftarrow m+1$

for $k \leftarrow 1$ to n

if $j > n$

$B[k] \leftarrow A[i]; i \leftarrow i+1$

else if $i > m$

$B[k] \leftarrow A[j]; j \leftarrow j+1$

else if $A[i] < A[j]$

$B[k] \leftarrow A[i]; i \leftarrow i+1$

else

$B[k] \leftarrow A[j]; j \leftarrow j+1$

for $k \leftarrow 1$ to n

$A[k] \leftarrow B[k]$

Figure 1.6. Mergesort

Recursion "tree":